



OPERATIONAL MANUAL

Differential Pressure Indicator

MODEL NO: DPI153-1X

VERSION: 1.0



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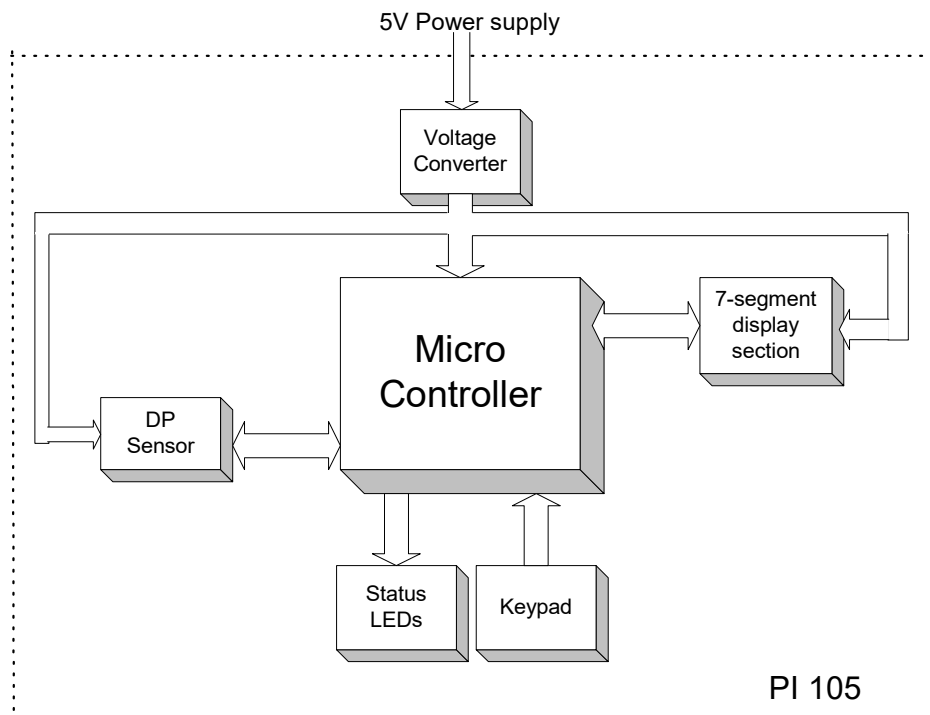
Description

The NEURAL “**Differential Pressure Indicator**” is specially designed to indicate the pressure difference between two different areas. The large 4 digit, 0.4” high LED display is easy to read, even from a distance. Two alarms can be selected for above or below set point control, which can help the user to monitor the DP and to take necessary actions to maintain the room conditions as per the requirement or application.

It has keypad to programming the instrument by user. It has Stainless steel Modular Face plate, which is designed as a mounting accessory for monitor suited for Modular wall Clean Room applications where you need flush mounting mechanism. This instrument enables you to measure humidity, temperature and differential pressure in the following fields:

- HVAC (Heating, ventilation, Air conditioning, cooling)
- Pharmacy industry.
- Blood stations, pharmacies.
- Horticultural and cultivation of plants.

2. Block Diagram



3. Features and Specifications

Features:

- Advanced Microcontroller based inbuilt DP measurement.
- Software programmable channel ranges, units & resolution.
- User programmable for measurement.
- 4 Digit, 0.39" bright LED display for displaying differential pressure.
- Process parameter status indication with LEDs.
- Process value violations display indication.
- User selectable set limits for process value violation.
- Inbuilt buzzer to audible process value violation.
- Password protection for programming mode.
- Software Calibration.

APPLICATIONS:

- Pharmaceutical industry.
- Blood Banks.
- Horticulture.
- HVAC (Heating, Ventilation, Air conditioning, Cooling).

4. Specifications:

DISPLAY:

Display Type : 4-digit .4" 7 segment LEDs.

ACCURACY:

± 1.5% of reading ± 3 Pa

± 1.5% of reading ± 3 mmH₂O

± 1.5% of reading ± 3 mbar

RESPONSE TIME:

0.3Sec

GENERAL

Operating Temperature Range: 0 to 50°C

Storage Temperature: -10 to +70°C

Environment: Air and Neutral gases

Enclosure: SS 316 face plate with MS Powder coated box.

Weight: 0.3Kg. (Approx.)

Size: 100X80X40mm (Approx.)

Face Plate: Flush mounting SS316 Face Plate.

Mounting: 1) Brick Wall Mounting.

2) Modular Wall Mounting.

POWER SUPPLY:

24Vdc, 250mA

CONSUMPTION:

2 Watts

Working Principle:

A Piezo resistive sensitive element creates a proportional voltage from the pressure applied on the sensor.

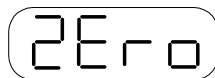
Units of Measurement & Ranges:

S. No.	Units	Range	Resolution
1	mmWC	-105.4 to 105.4	0.1
2	Mbar	-10.34 to 10.34	0.01
3	mmHg	-7.75 to 7.75	0.01
4	cmH ₂ O	-10.54 to 10.54	0.01
5	inch H ₂ O	-4.15 to 4.15	0.01
6	Psf (USA)	-21.6 to 21.6	0.1
7	hectopascal	-10.34 to 10.34	0.01
8	PSI	-0.150 to 0.150	0.001
9	Pascal	-1034 to 1034	1
10	Kilopascal	-1.034 to 1.034	0.001

5. Instrument Operation

Pressure Indicator has different modes of working. Description of each mode is given below:

- a. **Measurement mode:** Measurement mode is default mode for Instrument. In measurement mode instrument displays process value.
- b. **Configuration mode:** When PRG Key pressed in measurement mode causes to enter into configuration mode. In configuration mode, we can configure/change the several key parameters. .
- c. **Violation mode:** When process value exceeds the set limits range, Instrument will enter into violation mode measurement mode. In violation mode violated process parameter display section will starts flashing and respective status LED becomes glow in red color and buzzer also becomes ON.
- d. **Auto Zero Calibration mode:** Hold the OK key for 5 sec causes to enter into auto zero calibration mode from measurement mode. In this mode present differential pressure value will take it as zero offset. Every unit has its own zero offset value. This mode indicates with ZERO display in place of differential pressure display section. This mode remains for 5 sec only then it automatically comes back to measurement mode.



- e. **Unit display mode:** INC (◀) key press in measurement mode causes to enter into this mode. In this mode present measured pressure unit will display at pressure display section. This mode remains for 5 sec.



Display Indication of each unit is as mentioned below:

mmWC -> 

mbar	->	mbar
mmHg	->	mmHg
cmH2o	->	cmH2o
Inch. H2o	->	inH2o
Psf(USA)	->	PSF
hPa	->	hPa
psi	->	PSI
Pascal	->	PA
kPa	->	KPA

Figure: 1

6. Configuration mode:

To enter into this we need to enter valid **Password**. It will asks code reentry, if wrong password enters, this for once only. If enter wrong password for second time it automatically comes out to measurement mode. Default **Password** is 1111. User can change this in entry code change mode.

Configuration mode has multiple number of Programming parameters and sub modes to configure, As stated below:

- Unit selection
- Set point Range Selection
- Password Change

6.1. How to select mode: Valid password entry causes to display first programming parameter. Password reentry is allowed for once only, in case of wrong password entry. To wrap around the parameters selection, use INC/DEC key. To change the specific parameter, press OK key to accept it. Flow for one of the parameter or sub mode selection is as shown below.

S.No	Action to perform	Action to be Takes placed	Display
1.	To enter into configuration mode.	Press PRG key, then it will ask for password entry.	PSd
2.	Edit the password to 1111	Use ◀/▶ to edit the value and press OK to conform.	1111
3.	To accept the enter password and displays first programming parameter <i>Unit selection</i> mode.	Press OK key	Un it
4.	To go to <i>Set point range selection</i> mode	Press ▲ key	SEtL
5.	To go to <i>New Password selection</i> mode	Press ▲ key	nPSd

- Press OK key to edit the corresponding programming parameters mode. Then corresponding mode parameters will display and press ESC key to come out from the mode.
- If time out occurs, no key occurs for 30 sec, instrument automatically comes out from configuration mode and enters into measurement mode.

6.2. Individual mode Description: This section describes about how to edit each programming

Parameter mode and brief description about each one.

6.2.1 Unit Selection: This mode is to edit the unit of pressure and temperature parameters.

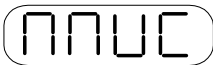
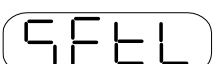
Note: When pressure units changes resolution of the parameter automatically changes to specified value in instrument specifications.

Instrument supported unit are as shown below table:

S.No	Process Parameter	Supported Units	Default Unit
1.	Differential Pressure	mmWC, mbar, mmHg, cmH ₂ O, Inch H ₂ O, psf(USA), psi, hPa, kPa, Pa	mmWC

Table: 1

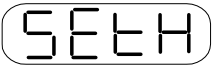
Operational flow to select is as shown below:

S.No	Description	Action to be Takes placed	Display
1.	Differential Pressure Unit selection: Present pressure unit will display, after accepting the temperature unit by press OK key.	Press OK key	
2.	To select required unit press INC or DEC key.	Press ▲key	
3.	To accept edited unit.	Press OK key	

6.2.1 Process Parameter Set point range Selection:

Every process parameter has low and high set point. By default these values are as same as Maximum values of that parameter specified in instrument specification.

S.No	Description	Action to be Takes placed	Display
1.	To enter set point selection press OK key.	Press OK key	

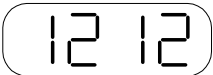
2.	Edit the value to required value. Editing digit appears with blinking. To editing digit will select through ◀ key.	Use ◀/▶ keys	
3.	To accept the edited value, press OK key. Then it will display Differential pressure High set point value.	Press OK key	
4.	Edit the value to required value. Editing digit appears with blinking. To editing digit will select through ◀ key.	Use ◀/▶ keys	
5.	To accept the edited value, press OK key. Then it will display next programming parameter Calibration	Press OK key	
6.	To accept the edited value, press OK key. Then it will display next programming parameter new password selection.	Use ▶ key	

6.2.2 New Password Selection:

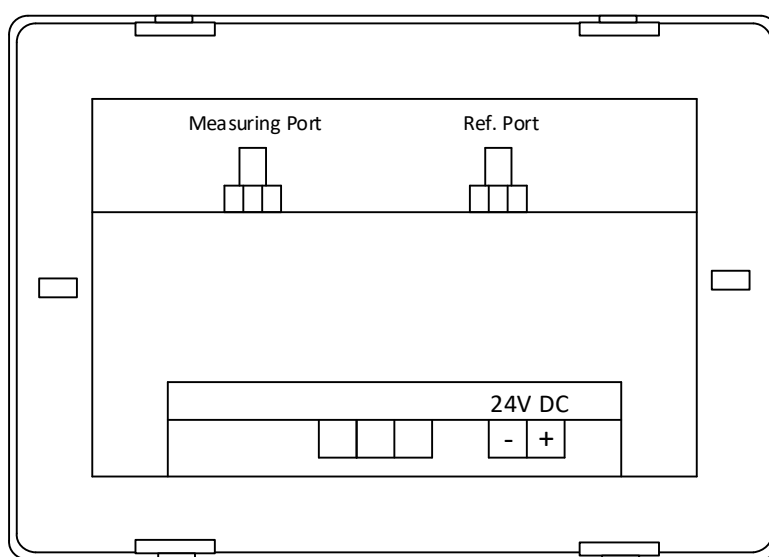
This password should enter to enter in to configuration mode to edit the programming parameters
Default code is 1111.

This describes how to change the password. From next time onwards user should enter new one.

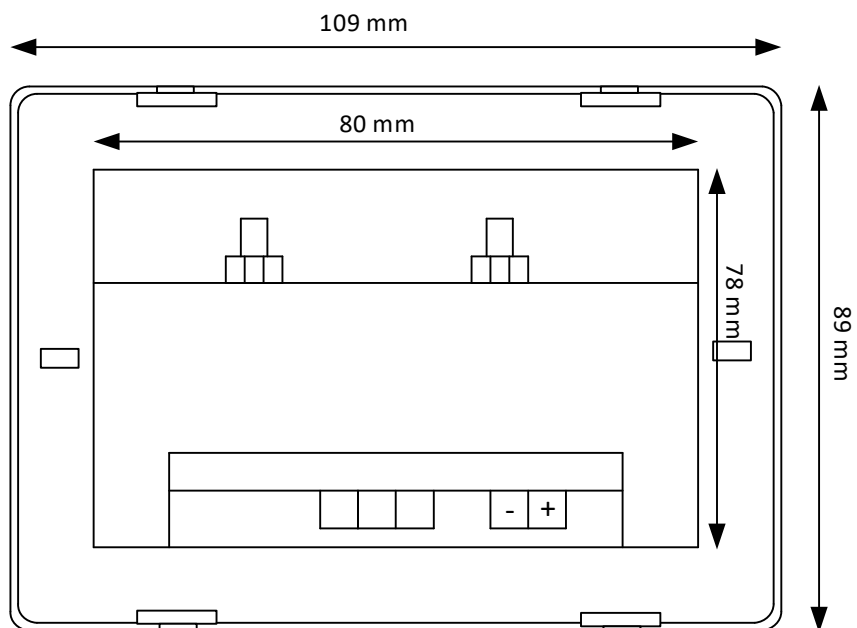
S.No	Description	Action to be Takes placed	Display
1.	<i>New Password:</i> After accepting the <i>New Password</i> by press OK key, it will display present configuration mode entry code. By default it is 1111.	Press OK key	

2.	Edit the value to required value. Editing digit appears with blinking. To editing digit will select through ◀ key. Press OK key to accept the edited value.	Use ◀/▶ keys	
3.	After press OK key it comes out from programming mode.	Press OK key	

7. Wiring Diagram:



8. Mechanical Drawing:



Cut Out Dimensions: 80 mm x 78 mm

9. Ordering Information:

I) DPI153 Indicator:

<i>Model No</i>	X	<i>Description</i>
DPI153	1X	RS485 MODBUS O/P
DPI153	2X	4 – 20 mA O/P

Notes:

For Further Information Contact:

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